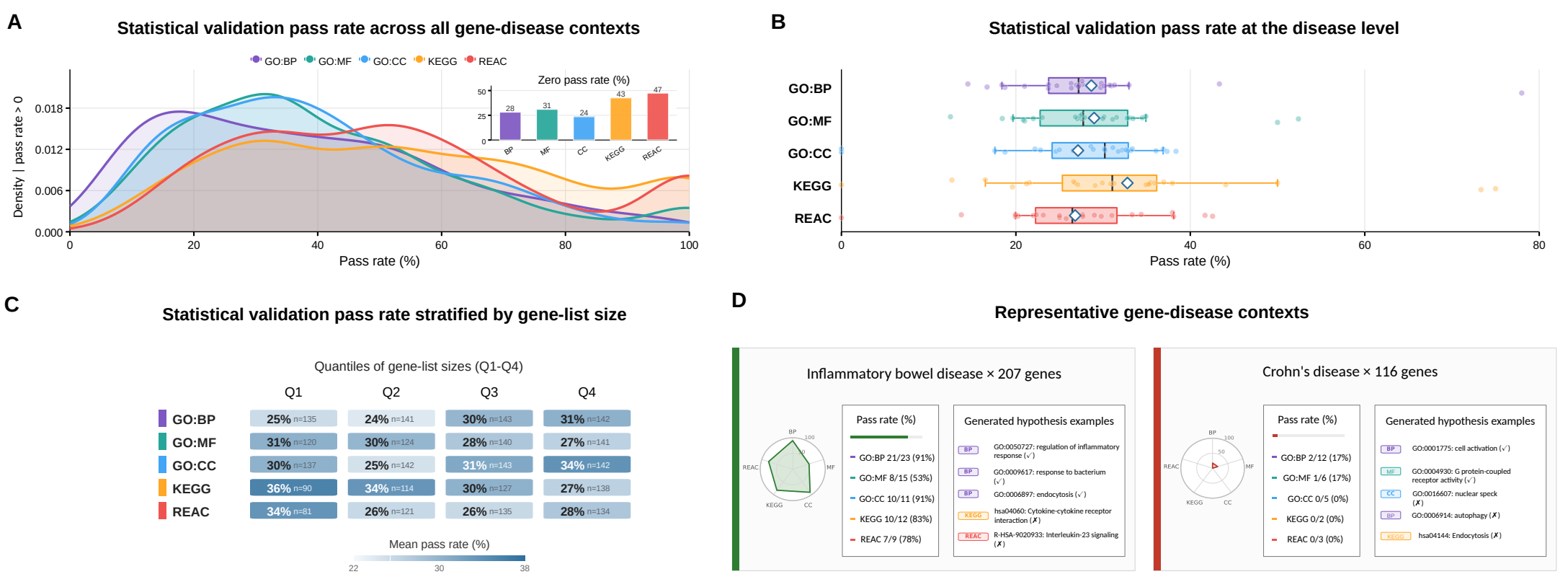

A
Statistical validation pass rate across all gene-disease contexts
GO:BP
GO:MF
GO:CC
KEGG
REAC
Zero pass rate (%)
28
31
24
43
47
BP
MF
CC
KEGG
REAC
Density | pass rate > 0
0.018
0.012
0.006
0.000
0
20
40
60
80
100
Pass rate (%)
B
Statistical validation pass rate at the disease level
GO:BP
GO:MF
GO:CC
KEGG
REAC
0
20
40
60
80
Pass rate (%)
C
Statistical validation pass rate stratified by gene-list size
Quantiles of gene-list sizes (Q1-Q4)
Q1
Q2
Q3
Q4
GO:BP
25% n=135
24% n=141
30% n=143
31% n=142
GO:MF
31% n=120
30% n=124
28% n=140
27% n=141
GO:CC
30% n=137
25% n=142
31% n=143
34% n=142
KEGG
36% n=90
34% n=114
30% n=127
27% n=138
REAC
34% n=81
26% n=121
26% n=135
28% n=134
Mean pass rate (%)
22
30
38
D
Representative gene-disease contexts
Inflammatory bowel disease × 207 genes
BP
MF
CC
KEGG
REAC
Pass rate (%)
GO:BP 21/23 (91%)
GO:MF 8/15 (53%)
GO:CC 10/11 (91%)
KEGG 10/12 (83%)
REAC 7/9 (78%)
Generated hypothesis examples
BP GO:0050727: regulation of inflammatory response (✓)
BP GO:0009617: response to bacterium (✓)
BP GO:0006897: endocytosis (✓)
KEGG hsa04060: Cytokine-cytokine receptor interaction (✗)
REAC R-HSA-9020933: Interleukin-23 signaling (✗)
Crohn's disease × 116 genes
BP
MF
CC
KEGG
REAC
Pass rate (%)
GO:BP 2/12 (17%)
GO:MF 1/6 (17%)
GO:CC 0/5 (0%)
KEGG 0/2 (0%)
REAC 0/3 (0%)
Generated hypothesis examples
BP GO:0001775: cell activation (✓)
MF GO:0004930: G protein-coupled receptor activity (✓)
CC GO:0016607: nuclear speck (✗)
BP GO:0006914: autophagy (✗)
KEGG hsa04144: Endocytosis (✗)